

Stimuli Panel Randomizer – User guide

1. General information

You can find the application on PNAS in: “S:\Partage\Electrophysiology\To perform electrophysiology\Stimuli Panel Randomizer”

To work, the full folder (~ 500 mo) needs to be copied somewhere on an internal or external drive. It seems that it doesn’t work from the PNAS.

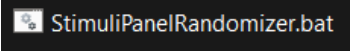
This application lets you randomize the order of a given panel of stimuli, with different quantities of the same stimulus in increasing order, with optional non-randomized leading and trailing stimuli panel.

Once started, the application will the user interface within a web browser.

The application can make use of Excel or tab-separated files (which are files similar to CSV / comma-separated files but with tabulation to delimit column separation).

2. Using the application

2.1. Starting the application

Double-click on the  shortcut placed on the desktop or the  file in the “StimuliPanelRandomizer” folder (See the orange arrow in the screenshot below).

When the application starts, on loading windows stating the availability of the packages is verified should appear for some time. Then the application should open in the default web browser.

2.2. Defining and saving a stimuli panel

The application considers up to 3 sections to the panel of stimuli:

- 1) “Controls”: A non-randomized leading stimuli panel.

The typical usage of this first part of the panel, is to be able to have negative or positive controls before proceeding to stimulation with your test panel.

2) “Test”: The randomized stimuli panel.

The part of the stimuli panel that will be randomized.

If multiple quantities of the same stimulus are present, they will be listed altogether in increasing order.

3) “Terminal”: A non-randomized trailing stimuli panel.

To be used if you need some stimuli to be presented last. Will not be randomized.

You can define the different stimuli in the different sections of the panel in two different manners.

2.2.1. Manually setting the stimuli panel

On the user interface, you will find blue buttons: ,  and . Clicking those, will add a line with two blank boxes to the respective section of the stimuli panel:   . The first box let you define the name of the stimuli and the second box the associated quantity. The  button let you delete the associated row.

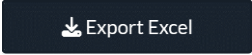
2.2.2. Importing a stimuli panel from a file

You can also click the  button in the “**Bulk Import / Export**” section to be prompted to upload an Excel file (.xls or .xlsx) or a tab-separated file (.tsv or .txt) that is listing your stimuli panel. These files must have 3 columns: “stimulus_type”, “stimulus” and “quantity_uq”. Respectively informing the software on: the section of the panel (control, test or terminal) to which belong the stimuli, the name of the stimuli and the quantity of the stimuli.

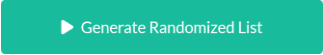
Uploading a file will fully replace the current panel in the user interface by thus contained in the file. After having uploaded a file, you can still modify the panel by using the  or  buttons.

An example .tsv file can be found in the “example files” folder: “Input_stimuli_panel-example.tsv”. You can use this file as a template to create a stimuli panel.

2.2.3. Exporting the defined stimuli panel

To avoid having to re-create a stimuli panel after closing the application, you can also export a file containing the information about the panel you create using the  or  buttons. By default, the file will be downloaded in the “Downloads” folder of the computer and the current date and time will be written in the name of the file.

2.3. Randomizing the panel

Once a panel is set, you can randomize the order of the “Test” section of the panel by clicking the  button. The randomized panel should appear on the right side of the user interface.

You can then export this randomized panel by clicking the ,  or  button, which will respectively 1) copy the generated sequence to the clipboard of your operating system to be pasted at your own discretion 2) export the generated sequence in a tab-separated file or 3) export the generated sequence in an Excel file.

By default, the file will be downloaded in the “Downloads” folder of the computer and the current date and time will be written in the name of the file.